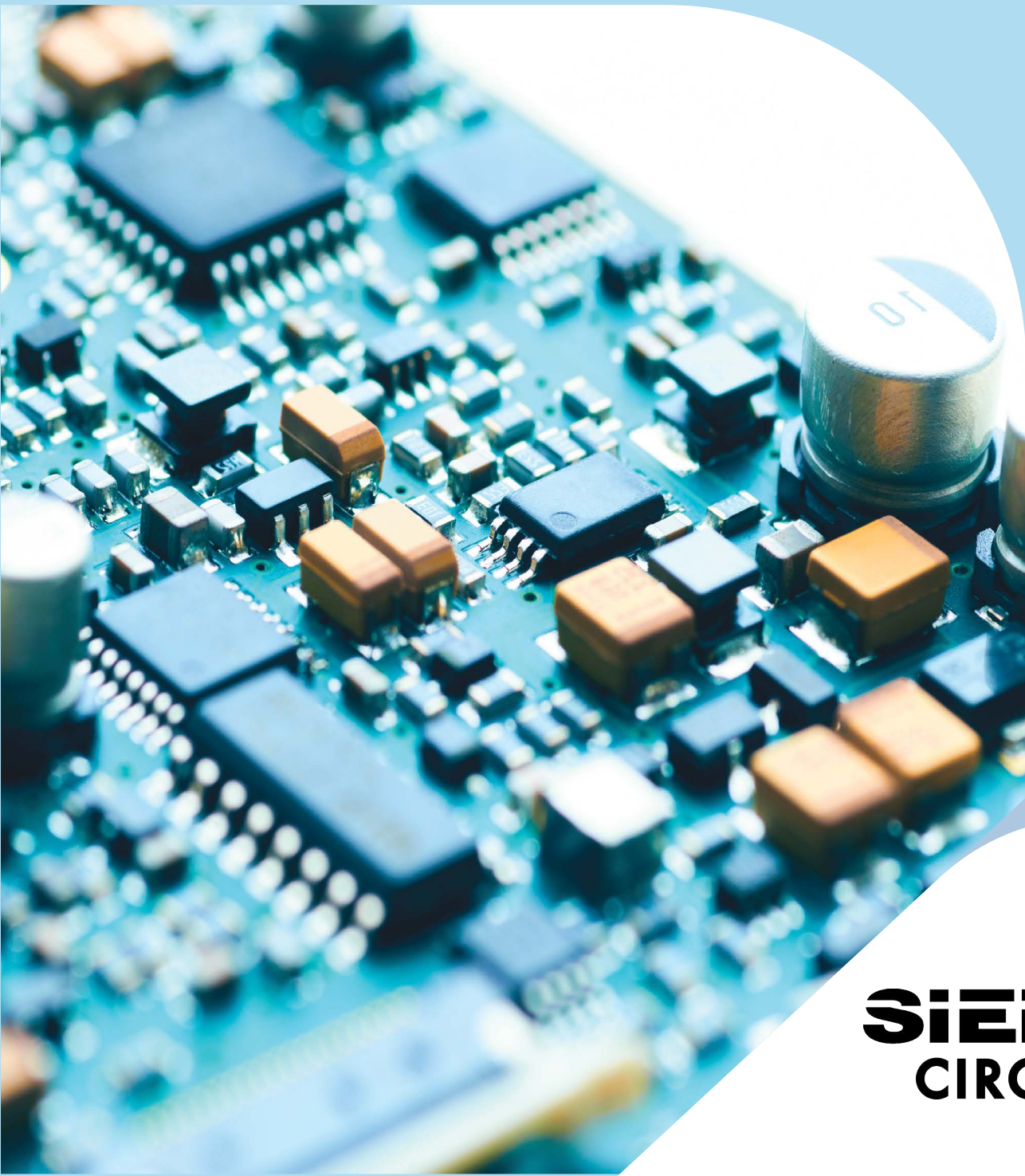


High-Speed Material Design Guide



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Table of Contents

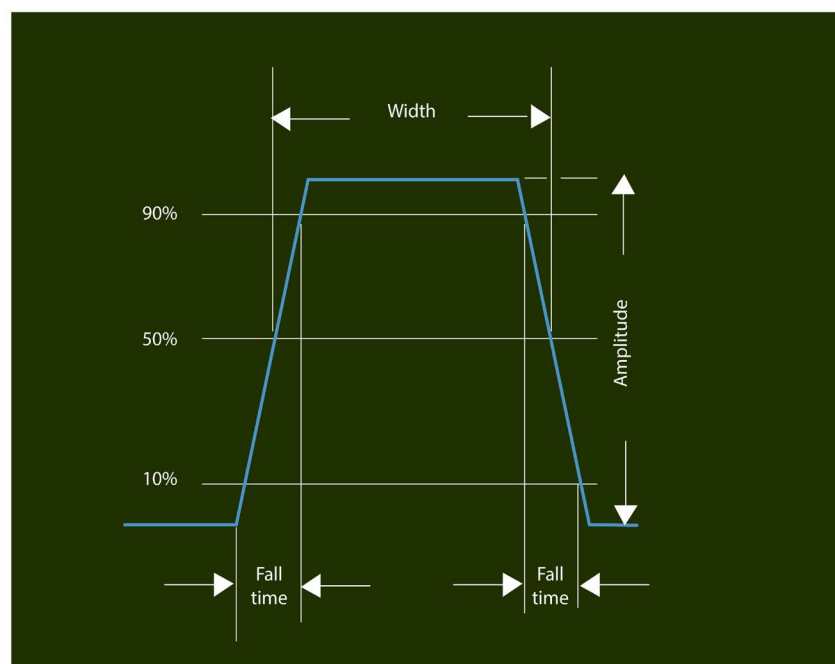
1. When do you define a signal as high-speed?.....	3
2. Why material selection needs special attention in high-speed design?.....	4
2.1 Signal loss and operating frequency.....	4
2.2 Variation of impedance due to the weaving of substrate.....	5
2.2.1 Mitigating fiber weave effect.....	6
3. Factors that affect high-speed material selection.....	7
3.1 Dielectric constant (Dk).....	7
3.2 Dissipation factor (Df).....	8
3.3 Thermal coefficients of Dk and Df.....	8
3.4 Coefficient of thermal expansion (CTE).....	8
3.5 Glass transition temperature (Tg).....	8
3.6 Decomposition temperature (Td).....	8
3.7 Thermal conductivity (k).....	9
4. Elements of high-speed PCB materials.....	9
4.1 Types of resin used in PCB laminate.....	9
4.1.1 Polytetrafluoroethylene PTFE.....	10
4.1.2 Polyimide.....	10
4.2 Types of glass cloth.....	10
4.2.1 Glass fiber construction (weave).....	10
4.2.2 Different types of glass.....	11
5. Popular high-speed materials.....	12
5.1 Rogers 4350B.....	13
5.2 RT Duroid 5880.....	14
5.3 Panasonic Megtron 6.....	14
5.4 Isola I-Speed material.....	14
6. High-speed material cost.....	15

1. When do you define a signal as high-speed?

A signal is defined as high-speed depending on the rise and fall time of the clock signal. Theoretically, they have broad bandwidth with frequencies that can extend to infinity. But, signals frequencies above 50 MHz are practically categorized as high-speed signals.

Usually, the clock signal is considered a square signal, but instantly interchanging between low and high levels is impossible. Hence, the square is replaced by a trapezoid in the time domain. Additionally, we consider rise time for calculations as the fall time extends longer due to the effect of transistors in the circuit. The rise time ranges between a few picoseconds to nanoseconds (typically 5 ns) for a high-speed signal.

In the frequency domain, the amplitude of higher frequency harmonics of the clock signal depends upon its rise and fall time. This means signals with a longer rise time will have a smaller magnitude of harmonics.



Trapezoidal clock signal

Being a board designer, you need to look into PCB stack-up, EMI, and power integrity to maintain the signal integrity in your high-speed designs. In this e-book, you will learn about the material properties and other parameters that should be considered for high-speed stack-ups.

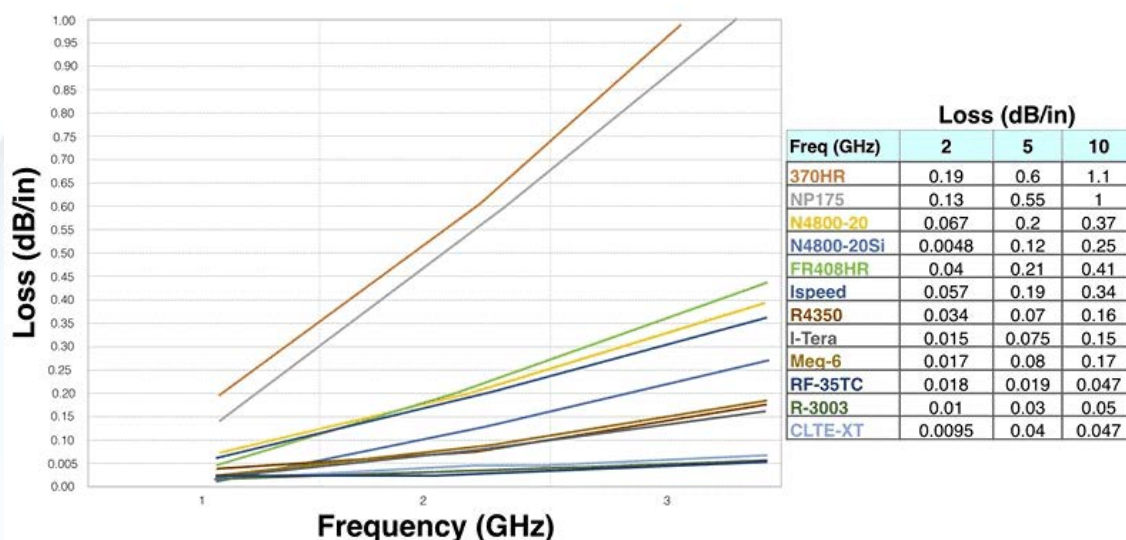
2. Why material selection needs special attention in high-speed design?

The parameters such as temperature rise, interconnect impedance, peel strength, signal, and power loss depend on the PCB material's characteristics. Hence, the chosen material's chemical, electrical, and thermal properties will significantly affect the functioning of the PCB.

The main factors to consider while selecting the ideal base substrate for your high-speed design are mentioned below:

- 1) Signal loss at the operating frequency
- 2) Weave effect of the substrate

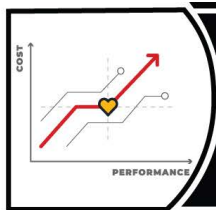
2.1. Signal loss and operating frequency



Plot of loss v/s frequency of different materials

When the digital trapezoidal signal propagates through the material, it experiences attenuation and pulse broadening, hence signal quality reduces. It is evident from the above graph that signal loss in the material increases as the frequency increases. The proportion at which it increases varies in different kinds of material. From the table, we can conclude that Rogers-3003 is a better choice for high-speed design than FR 408HR.

Apart from this, other signal integrity issues such as cross talk, EMI, ground bounce, etc., will become a severe problem at higher frequencies. Therefore, select a suitable material that is stable at high frequency.



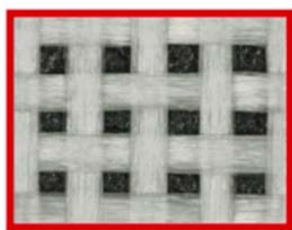
PCB Material Selector

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2.2 Variation of impedance due to the substrate weave

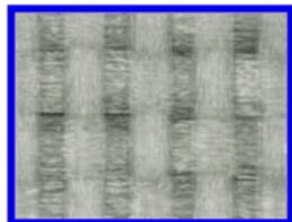
As you already know, the substrate/prepreg used in a PCB is made of bundles of glass fibers woven and hardened using resin.

Loose Weave



Dk distribution is *uneven*.
Thickness of the board varies widely.

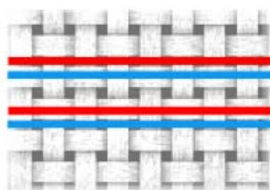
Tight Weave



Dk distribution is even.
Thickness of the board does not vary.

Weave of glass fiber and Dk distribution

The arrangement of the glass fiber has a high impact on the dielectric constant of the material. It also determines the impedance characteristics. Hence, a material with even dielectric properties will have a uniform characteristic impedance which is essential for signal integrity.

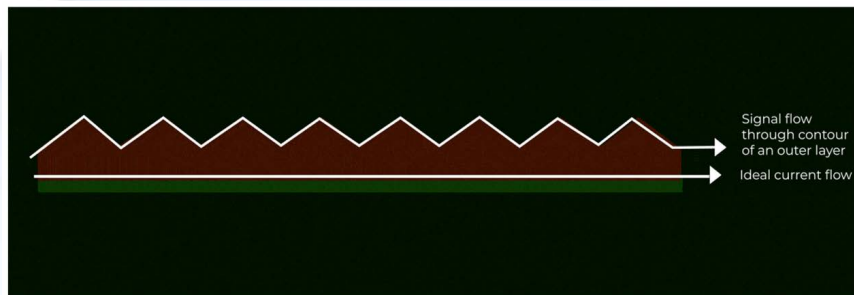


Signal transmission
on glass weave

You can observe two signals propagating through the laminate in the image above. Typically, glass fiber has a higher Dk value, whereas resin has a lower Dk value. When the signal (red) flows mainly on the glass fiber, it will have a constant dielectric value. In contrast, when the signal (blue) flows through the region alternating between fiber and resin, a rapid change in the Dk value will affect the uniform impedance, and this causes signal distortion.

This signal skew becomes significant considering the board's entire stack-up at a higher frequency. Therefore, a denser and closely spaced weave of glass fiber is preferred for high-speed design. The variation in impedance due to the glass weave is called fiber weave effect. The same principle applies to copper surface (copper and dielectric interface).

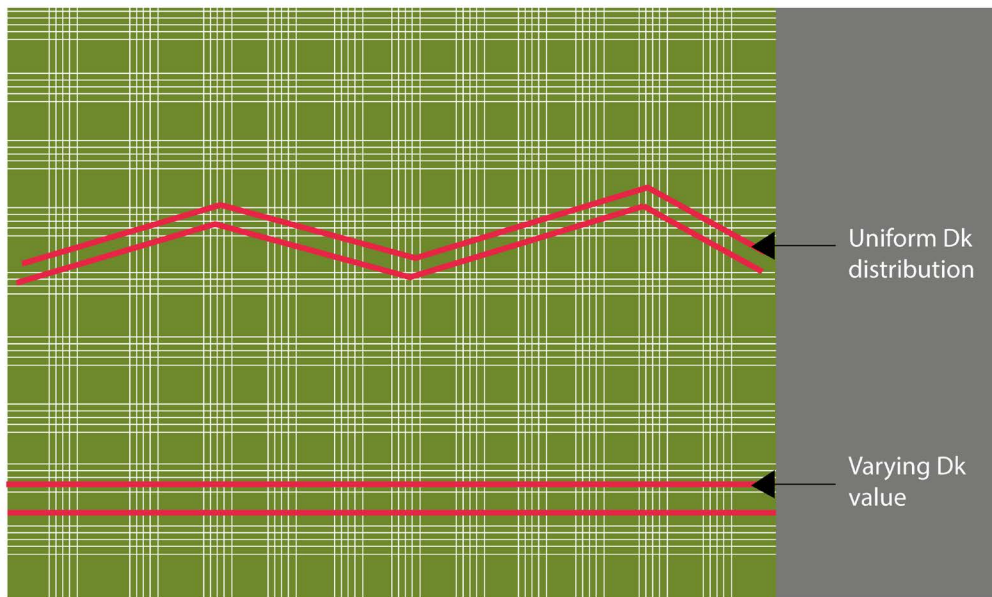
Smoother the copper surface, higher the signal integrity. See the image below. An interesting thing that's observed here, in higher profile the signal will have to travel along contours maximizing its signal travel path. This adversely affects the signal integrity. Hence choose low profile for high-speed designs.



Signal flow on copper surface

2.2.1 Mitigating fiber weave effect

- Select a fiberglass weave style that is more evenly distributed across the board, such as 2116 and 3113.
- Choose a material whose resin and fiberglass have closer/similar dielectric constant.
- While designing the circuit, use wider trace as practically possible and route them at an angle with respect to wrap or fill.
- Implement zigzag routing for lengthy differential signals.



Zigzag routing to mitigate fiber weave effect

$$W = (I_{RMS}^2 \Delta T) / Th$$

$+$
 $\div$
 $\times$
 $=$

IPC-2152

Trace Width, Current Capacity and Temperature Rise Calculator

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3. Factors that affect high-speed material selection

A few terms are widely used across the industry when selecting the laminate material. The prominent material characteristics that affect the performance of a circuit board are discussed below.

3.1 Dielectric constant (Dk)

The textbook definition of dielectric constant (Dk) is the ratio of the charge stored in an insulating material placed between two metallic plates to the charge stored when the same material is replaced by vacuum/air. Generally, the dielectric constant indicates the ability of the material to store electrical energy.

As discussed earlier, the Dk value may vary throughout the material. The typical dielectric constant values mentioned in the datasheets are generally the average Dk value across the material. Considering dielectric constant is crucial for impedance and signal integrity. Typically, the value ranges between **2.5 to 4.5**.

3.2 Dissipation factor (Df)

The Df, also known as loss tangent ($\tan \delta$), is the tangent of the phase angle between the resistive and reactive current in the dielectric. It is the ratio of the imaginary and real parts of the dielectric constant. Generally, its value ranges from **0.001 to 0.030**. The dielectric loss is directly proportional dissipation factor which means that higher Df values result in higher dielectric loss.

Attenuation or rejection?

We need to draw a distinction between “real” loss (attenuation) and mismatch loss (rejection). Rejection usually means reflection. The rejected signal is returned to the generator where it is dissipated. Attenuation is the signal that is dissipated by the conductor and dielectric, generally as heat. Both “real” loss, and mismatch loss affect bandwidth but have separate causes, effects, and solutions.

3.3 Thermal coefficients of Dk and Df

The Dk and Df discussed above also have temperature-based variants called thermal coefficient of dielectric constant (TCDk) and thermal coefficient of dissipation factor (TCDf). They indicate the change in the material's Dk and Df values with respect to temperature variation. The lesser the change in the Dk and Df value, the more stable is the material with respect to change in temperature.

3.4 Coefficient of thermal expansion (CTE)

CTE measures the change in the object's size with respect to the variation in the temperature. The coefficient of thermal expansion is calculated in parts per million expanded for every degree Celsius change in temperature.

During the stack-up preparation, the PCB materials are heated up, so the CTE value is an essential parameter to consider while selecting the material.

3.5 Glass transition temperature (Tg)

Tg is the temperature at which the substrate transitions from a rigid to molten state. After reaching the Tg, the CTE value increases drastically. Hence, the CTE value above the Tg is larger than the CTE value below Tg.

Glass transition temperature is expressed in degree Celsius. For example, the Tg value of Rogers 4340 B is 280°C.

3.6 Decomposition temperature (Td)

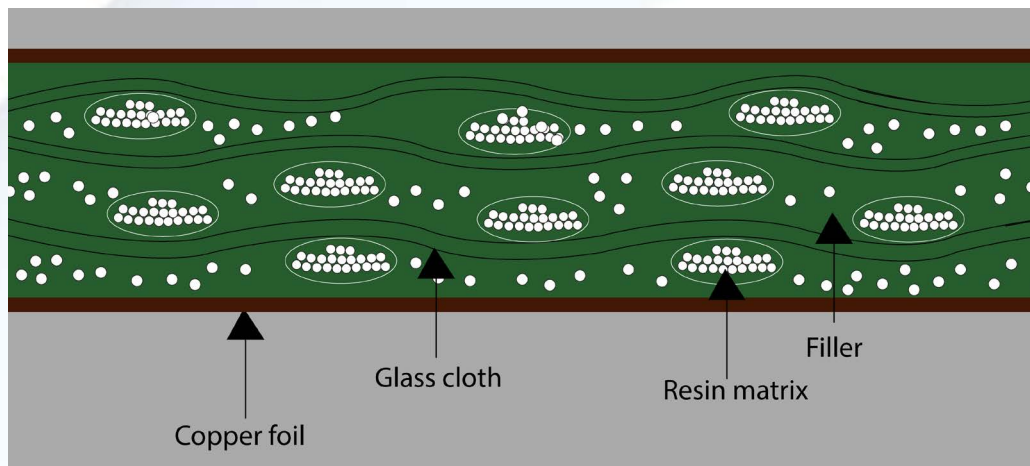
Similar to Tg, Td refers to a temperature at which the substrate decomposes chemically. This is also expressed in degree Celsius. The Td value of Rogers 4340 B is 390°C.

3.7 Thermal conductivity (k)

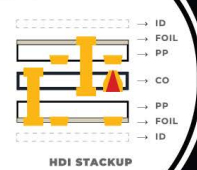
Thermal conductivity indicates the rate at which heat is dissipated from the substrate during operation. It is measured in Watts/meter Kelvin. The lower the value of k, lower the heat transfer, while the higher value, higher will be the heat transfer. This also determines the temperature rise with respect to ambient temperature. Thermal conductivity plays an important role in thermal management of a circuit board.

4. Elements of high-speed PCB materials

In a multilayer stack-up, alternate prepreg layers are stacked between two copper foils. It is best not to use more than 3 types of prepreps. Apart from the glass fiber weave and resin, the CB laminate can include fillers, reinforcement, and metal foil. Ensure the dielectric thickness of the prepreg is less than 10 mils to reduce the possibilities of variation in the final thickness.



A typical laminate structure



PCB Stackup Designer

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4.1 Types of resin used in PCB laminate

Thermosetting resins: These are the polymers that are initially in a liquid state and later harden through irreversible curing. Epoxy is a thermosetting resin that is used in majority of high-speed designs. Other resin materials include BT, PPE, cyanate ester, and modified acrylates.

Thermoplastic resins: These plastics are moldable at high temperature and solidifies at lower temperature. Polyimide and polytetrafluoroethylene are usually used in board construction.

4.1.1 Polytetrafluoroethylene (PTFE)

PTFE or popularly known as Teflon, is a thermoplastic fluoropolymer. It has a higher Tg that ranges from 160-280°C. They have excellent dielectric properties at high frequencies, which is ideal for high-speed designs. The thermosetting version of this is flexible and is supplied in a film form. It is mainly used in rigid-flex circuits. However, when compared to epoxy, they are a bit expensive.

4.1.2 Polyimide

Polyimide has good flexibility, electrical properties, chemical resistance, and tensile strength. It is the best choice for flexible PCBs. However, they can also be used in high-power, high-speed, and high-frequency applications.

4.2 Types of glass cloth

The selection of glass cloth is dependent on two factors

- Type of weave (glass fiber construction)
- Kind of glass utilized

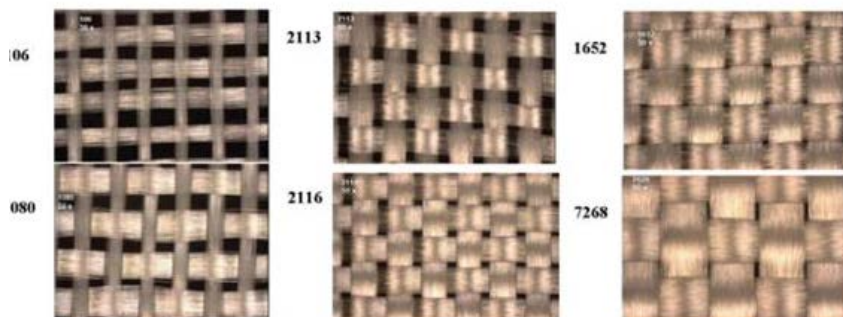
Note: Presently, there are “cloth less” laminates and “filled” laminates that will help reduce or eliminate weave problems.

4.2.1 Glass fiber construction (weave)

As explained in the previous sections, different fiberglass arrangements will impact the Dk and impedance values. A loose weave results in higher variation, and a tight wave produces lower variation in Dk distribution. The other advantages of tight glass fiber weave are:

- The thickness of the board will be uniform
- Drilling holes will be consistent and efficient
- Easier to manufacture
- The average Dk remains consistent across the board

However, too much low resin content can cause resin starvation during lamination. So avoid using too much closely spaced glass fiber construction. For example, 2116 and 7628 laminations have low-resin and high-glass content.



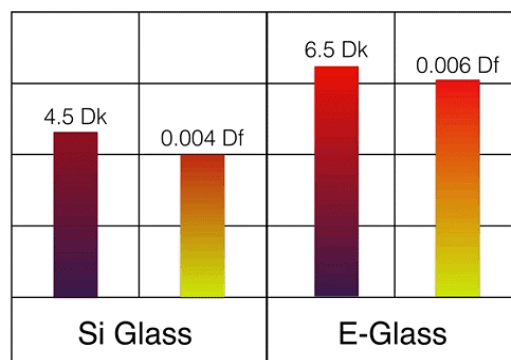
Different types of fiberglass weave in materials

Glass Style	Count Warp	Count per Inch	Warp Yarn	Yarn Fill	Glass Thickness
106	56	56	ECD-900 1/0	ECD-900 1/0	0.0015
1080	60	47	ECD-450 1/0	ECD-450 1/0	0.0025
1280	60	60	ECD-450 1/0	ECD-450 1/0	0.0022
2113	60	56	ECE-225 1/0	ECD-450 1/0	0.0029
2116	60	58	ECE-225 1/0	ECE-225 1/0	0.0038
3070	70	70	ECDE-300 1/0	ECDE-300 1/0	0.0034
1652	52	52	ECG-150 1/0	ECG-150 1/0	0.0045
7628	44	31	ECG-75 1/0	ECG-75 1/0	0.0068

4.2.2 Different types of glass

There are two kinds of glasses to choose from E-Glass and Si Glass.

Si Glass versus E-Glass at 5 Ghz

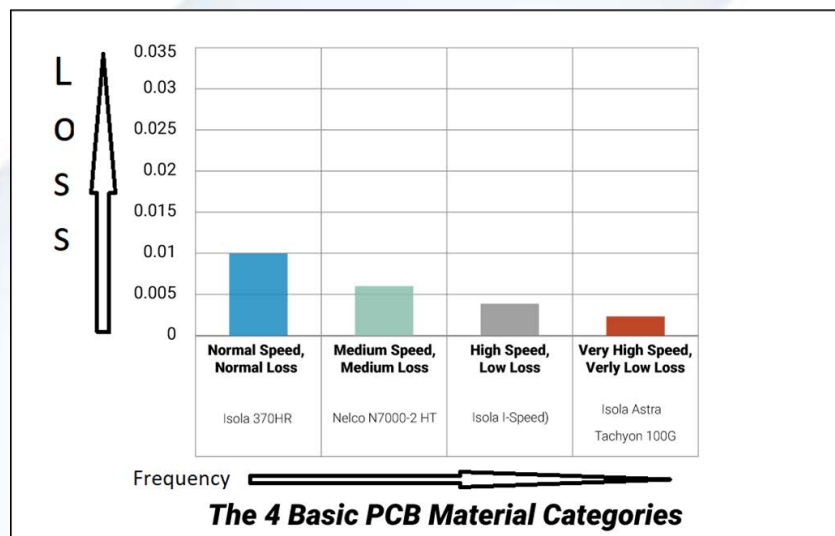


Si glass v/s E-glass at 5 GHz

E-glass: This is a low alkali glass with a typical nominal composition of SiO₂ 54wt%, Al₂O₃ 14wt%, CaO+MgO 22wt%, B₂O₃ 10wt%, and Na₂O+K₂O less than 2wt%. The majority of PCBs use E-glass in their construction. It has some impurities, which means other materials may also be present in the composition. This has high Dk (6.5) and Df (.006) values at 5 GHz frequencies compared to the SI glass. Its thickness varies from 1.3 mils to 6.8 mils.

Si glass: The purity of Si glass is more when compared to E-glass. As a result, its Dk (4.5) and Df (.004) values at 5 GHz are lower and cost 15% more than the E-glass.

5. Popular high-speed materials



Plot of loss v/s material categories

Signal frequency below 500 MHz

Application: medium speed, normal loss

- FR370HR (Isola)
- N7000-2 HT (Nelco)

Signal frequency 500 MHz to 3 GHz

Application: high speed, low loss

- FR408HR (Isola)
- I-Speed (Isola)
- Megtron6 R-5775 (Panasonic)

Signal frequency 3 GHz and above

Application: very high speed/frequency, very low loss

- I-Tera MT40 (Isola)
- RO3003 (Rogers)
- RO4350 B (Isola)
- Tachyon-100G (Isola)
- Astra MT77 (Isola)

5.1 Rogers 4350B

This is predominantly used for high-speed circuits. There are three prepreg choices in Rogers 4350B laminates:

- 4 mil prepreg with two glass styles
- 3.6 mil prepreg with one glass style

They are typically ceramic combined woven glass. Rogers is smooth and flat, which helps with impedance control. Hence, it provides low signal and dielectric loss. However, the manufacturing process of this material is more complex as it requires higher pressure for proper lamination, and the material cost is more when compared to Megtron 6.

- Rogers 4350B complies with IPC-4103 standards and RoHS regulations.
- It is CAF resistance.
- Tg and Td value is 280°C and 390°C, respectively.
- Thermal conductivity of the material is 0.69 (W/mk).
- Dielectric constant is 3.48 at 10 GHz.
- Dissipation factor is 0.002 at 10 GHz.

Rogers materials have low CTE and outgassing defects which is more suitable for advanced applications. The CTE values at different axis are mentioned in the table below.

Axis	CTE (ppm/ degree Celsius)
X/Y upper limit	12
X/Y lower limit	10
Z post Tg	32
Z pre Tg	32

5.2 RT Duroid 5880

Rogers RT/duroid 5880 is a high frequency laminates which has PTFE composites reinforced with glass. They have the low dielectric losses and dielectric constant. Therefore, it has uniform electrical properties over a wide range of frequency. Other features include:

- Chemical resistant
- Low moisture absorption
- Dk at 10 GHz is 2.2
- DF at 10 GHz is 0.0009
- Thermal conductivity is 0.20 W/m/K
- Td is 500°C

5.3 Panasonic Megtron 6

Megtron 6 laminates are similar to conventional FR-4, and the manufacturing process has no change in pressure, temperature, or cure time. The laminates come with three kinds of resin content with various prepreg glass styles. The resin is evenly spread across the surface of the tight weave, so there is a low change in impedance which is suitable for high-speed applications.

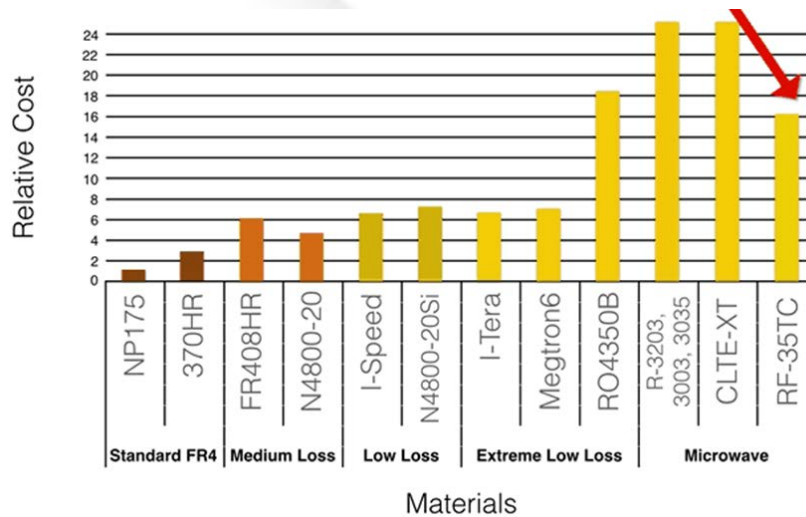
- The dissipation factor for the material is 0.002.
- Depending on the thickness, the dielectric constant ranges from 3.46 to 4.07.
- It is CFA-resistant.
- Tg measured through dynamic mechanical analysis is 210°C.
- CTE (Z axis) of the material below Tg is 45.

5.4 Isola I-Speed material

I-Speed material is a low-loss material used in high-speed and high-density interconnection (HDI) applications. It is compatible with RoHS, UL, and IPC standards. Its manufacturing process is similar to FR-4.

- Tg and Td are 180°C and 360°C, respectively.
- Dk is 3.64, and Df is 0.0060
- Thermal conductivity is 0.4 (W/mk)

6. High-speed material cost



Graph of material versus relative cost

One of the driving costs of manufacturing circuit boards is the material used for the layer build-up. High-speed, low-loss materials are a bit pricy. Selecting the best suitable and cost-efficient material is the key. For example, Taconic RF-35 is 30% less expensive than other materials in the microwave category and provides the same performance.

About Sierra Circuits

Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support which helps in producing an unprecedented board quality with high reliability. By choosing us you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Talk to our experts

Sierra Circuits helps PCB designers plan it right!

Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view.

Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Upload your data and receive a free consultation and review of your design. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Sierra Circuits PCB design capabilities

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
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- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via and micro vias, via stacking and staggering technology
- Designs with fine pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Stack-up design and impedance calculation for single-ended, differential, and coplanar-waveguide models

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IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D - military and aerospace
- ISO 13485:2016 certificate - medical devices

Mil-Spec certified and DLA approved

- MIL-PRF-31032/ 3 flex certificate
- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid-flex PCBs
- Rigid PCBs

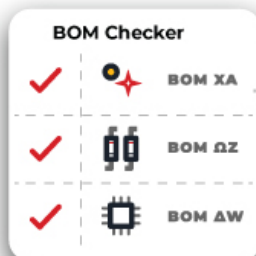
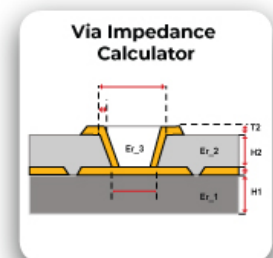
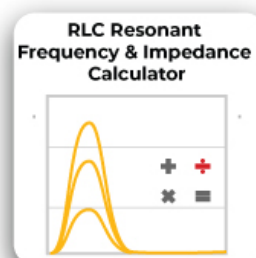
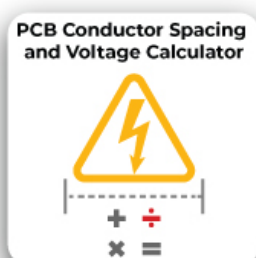
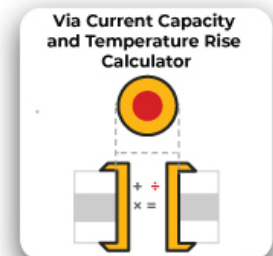
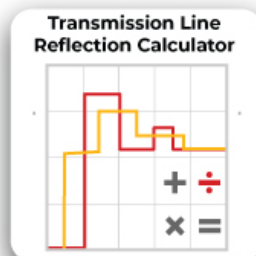
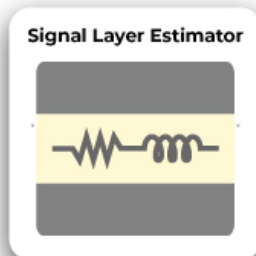
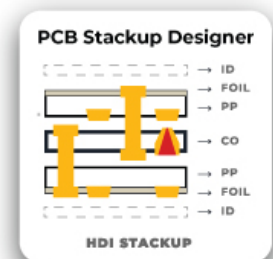
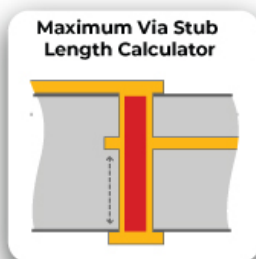
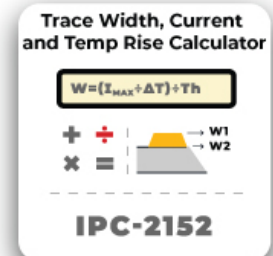
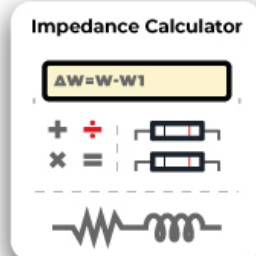
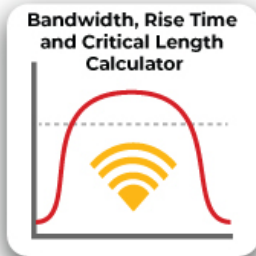
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